

MRI CASES CONTAINING CEREBRAL TUMORS RETRIEVAL USING BAYESIAN NETWORKS

Hedi YAZID¹, Karim KALTI¹, Fatma ELOUNI², Najoua BEN AMARA ESSOUKRI¹ and Kalthoum TLILI²

¹Engineers National school Sousse, Tunisia

e-mail: hedi.yazid@gmail.com, karim.kalti@gmail.com, najoua.benamara@eniso.mu.tn

²Radiology Departement, Sahloul Hospital, Sousse, Tunisia

e-mail: elounichedia@yahoo.fr, kalthoum.tlili@rns.tn

Abstract— We propose in this paper a Bayesian model for the retrieving of MRI (magnetic resonance imaging) exams that contain cerebral tumors. Bayesian network proved its efficiency and reliability in several AI (Artificial Intelligence) problems and especially in aid-decision applications. To diagnose a cerebral tumor in a MRI exam, we need to interpret diverse sequences and to refer to visual descriptors and, also, to the patient's clinical information's (age, sex, other diseases ...etc.). Our main idea is argued by the probabilistic aspect chosen in the decision making of diagnosis process. This aspect will be translated as a probabilistic decision model. Our work is tested in a several medical cases that were collected from Sahloul Hospital. Performance indices of experiments are promising.

Keywords-component; *Bayesian network, Retrieval, Indexing, MR Imaging, Cerebral tumors, Similarity Measurement, Euclidian Distance.*

I. INTRODUCTION

The daily increasing evolution of the images number handled in various domains, conducts to the creation of more convenient structured images databases. This fact supported the development of various applications such as filing, the search and the extraction for relevant information's in a multi-media flood. The image indexing is a more and more requested axis. It's worthy to note that the increasing evolution of the use of images leads to an essential development of indexing algorithms. Similarity-based retrieval has become an important area of computer vision research. It has reached a state of maturity in that it is starting to be used in real commercial and medical applications. [14].

The medical domain is an ideal field of indexing application. Medical images are produced in an incremental way. The current systems do not offer an intelligent research in the exponential files of hospitals. However, the archived medical cases can be re-used in the aid-diagnosis process, in the scientific research and in the tracking of files' treatment. In this occurrence, various are the contributions in the indexing of the cerebral tumors. [14], [2] [17].

Studies concerning the medical images retrieval are varied. We mention contributions using the physical characteristics such as color, form, texture... etc [4], [17], other contributions based on the semantic characteristics [18] and works combining the two types [7]. The combination of the high level characteristics (semantic) and

the low level ones (visual) is the most applied choice in recent studies.

This paper proposes a model for a retrieving framework of MRI cases containing cerebral tumors. In the actual work, we focus on the similarity measurement and the decision step. The model is describing the decision-making process of a radiologist facing a tumor. We meet this objective by reasoning in a Bayesian network. The similarity quantification algorithm primarily benefits from the contribution of the Bayesian inference while treatment such problems.

This paper is organized as follows. In section 2, we cite some related works that apply Bayesian networks in retrieval contributions. In section 3, we review Bayesian networks. Next, in section 4, we present the study framework of this work. In section 5, we describe in detail our contribution and we propose the model. In section 6, we quantify the performance of our contribution by experiments. Finally, a discussion and a conclusion of the whole paper will be given.

II. RELATED WORKS

Many different techniques have been introduced for searching medical images in order to retrieve, from the image management systems, images and records that have similar if not exact content [20].

From this field research, we are going to focus on CBIR systems applied in medical applications. There are several works on this context. We mention, as illustration, the automatic search and selection engine with retrieval tools (ASSERT) which is a Content Based Retrieval system for high-resolution computed tomographic (HRCT) lung images database. This framework apply the human –in – the – loop approach in which the human delineates the pathology bearing regions (PBR) and a set of anatomical landmarks in the image [1]. We cite, also, the Content Based image retrieval for Medical Applications (IRMA) which is an approach for content – based image retrieval in medical applications with the particular focus on its semantic layers of information modeling, its hierarchical concept for feature representation, and its distributed system architecture for efficient implementation [18].

Recently, several contributions apply Bayesian networks in the modeling of images retrieval systems. We cite, as illustration, the Bayesian CBIR system Pichunter developed by Cox et al. [8]. Lio et al. proposed a framework for semantic image understanding [11]. Also, Bayesian approach is applied in multi feature based image retrieval in Zhang et al. [16]. From his part, Quellec et al. [6] used Bayesian networks for fusion information in Multimodal medical case retrieval.

III. BAYESIAN NETWORKS

The probabilistic graphic models, and more precisely the Bayesian networks [15], initiated by Judea Pearl in the 1980, are a very practical tools for the representation of uncertain knowledge and the reasoning based on incomplete information, in many fields like bio-data processing, risk management, marketing, the computer security, transport, etc. Bayesian network is a directed acyclic graph composed by inter-connected nodes; each node represents well defined information. [15]

However, building a BN (Bayesian Networks) is usually more difficult than building an ES because a BN contains a lot of additional quantitative (numeric) information which is essential to its proper operation. The Bayesian networks learning accomplish this task and, it is so, an indispensable axis in the researches applied on Bayesian networks.

A. Bayesian Networks Learning

Bayesian networks learning consist of calculating the probability of each hypothesis, given the data, and makes predictions on that basis.

That is, the predictions are made by using all the hypotheses, weighted by their probabilities, rather than by using just a single “best” hypothesis. In this way, learning is reduced to probabilistic inference. [12]

The purpose of the training is to estimate, starting from a database and a priori knowledge:

- The Bayesian network structure (X relative of Y?)
- The Bayesian network parameters ($P(X = 0 | Y = 1)?$), [10].

On the one hand, we can apply parameters learning in a Bayesian network and this concerns all Network nodes. After an exercise of parameters learning, the conditional probability tables (CPT) could be constructed from empirical evidences. On the other hand, we call structure learning when the structure of the Bayesian network is unknown. We apply in this case a heuristic search which looks for “good” structures.

B. Bayesian Networks Inference

According to the Bayesian rule, the posterior probability can be expressed by the joint probability, which can be further expressed by the conditional and prior probabilities.

Given a set of random variables $Y = \{Y1, Y2, \dots, Yn\}$ and the joint probability distribution $P(Y)$, a Bayesian network encodes the conditional independences which exist in the

joint probability, and which expresses in a factorized form. [9]:

$$P(y) = \prod_i^n P(y_i | Pa(y_i)) \quad (1)$$

The probabilistic inference problem, in particular Bayesian inference, consists in calculating a marginal or conditional probability of a sub –set variables or to identify the most probable realization of a hidden variables (not observed) knowing observed information’s.

IV. STUDY FRAMEWORK

The study data base contains MRI Cases collected from the MRI center in Sahloul Hospital. All cases involve an approved brain tumor interpreted by radiologists. A brain tumor is an intracranial solid neoplasm and which is included inside the cranium or in the central spinal canal [13]. The diagnosis of a tumor falls under a precise and complex clinical step leading thereafter to an adapted therapeutic decision. It is composed of several stages: suspicion, detection, observation, determination of its histological nature.

The World Health Organization (WHO) set up a nomenclature of the central nervous system tumors with an aim of succeeding, in the long term, with an international standard [13]. It is then; the cerebral tumors present a large histological variety. In spite of this polymorphism, their diagnostic approach is largely facilitated by the existence of a preferential distribution according to their localization and the age of patient.

Usually, we classify the cerebral tumors in:

- Intra-axial tumors developed starting from cerebral tissue,
- Extra-axial tumors developed in under-arachnoïdiens spaces (primarily starting from the Meningeal envelope) or in the bony wall of the cranial cavity.

In this work, we suggest to limit the study field on only one tumor family. The experts recommended us to focus on tumors belonging to the intra-axial SupraTentorial tumors family. The argument is that characteristic’s interpretation is too similar for the whole of the selected tumors. It is, thus, six tumors are concerned and which are the following ones:

TABLE I. CEREBRAL TUMORS INCLUDED IN THE SIMULATION.

1	Metastasis
2	Pilocytic Astrocytoma
3	Glioblastoma
4	Oligodendroglioma
5	Lymphoma
6	Low grade glioma

V. BAYESIAN BASED RETRIEVAL MODEL

A. Network Construction

In our proposed model, the characteristics (possible to occur) can be model as nodes. The conceived network is a fusion of all visual characteristics with semantic information on which the interpretation of a radiologist is based.

The network is composed of 3 levels: the first one, called “observed layer” representing variables corresponding to characteristics used in the interpretation process. The second one; called “the intermediate layer”; play the role of state nodes in a subset of the characteristics. For instance, the node “State of the tumor” which brings together the nodes “Limits Tumor”, “Size Tumor”, “Tumor Number”, “Seat Tumor” and “Localization Tumor”. Finally, a result node is integrated in the network and it deals with the task of decision « this current case is it belonging to a tumor c?».

As an illustration example, we present below a sub graph which contains nodes extracted from the retrieval based Bayesian Graph:

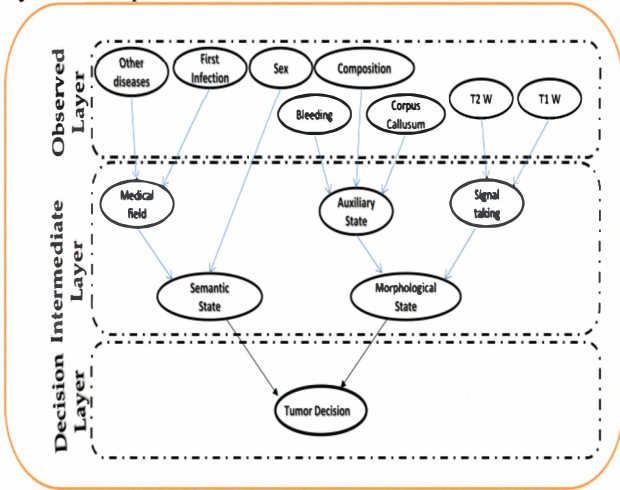


Figure 1. Cerebral tumor detection Bayesian network architecture

B. Similarity Measurement

The similarity estimation method is often the most influential task on the precision of an indexation system. Content-based image retrieval (CBIR) aims to return images which, according to human perception, are similar to a query image. Remarkably, few such systems consider what similarity means in the context of human usage. [3].

At this stage, our proposition has to investigate the contribution of the probability reasoning established by the application of the bayesian network. The algorithm requires two stages:

- A first comparison between the query case and the tumor's classes already initialized in the network. We deduce a membership vector of the current case with each tumor. The membership measurement is a probability.
- A second comparison between the vectors of membership of each case from the data base cases and the vector of the current case.

To formulate our problem, let C_r be a case suggested as being a request to be infiltrated in the network. To schematize our case, we define a set D_j , $j = (1, \dots, N)$ representing variables. Each characteristic is represented by a finite set of states $k = \{1 \dots K\}$. We will repeat this graph affectionation on the tumors list mentioned above and defined as $c = \{1 \dots C\}$. We propose C_i a treated case from the database. To measure the similarity between C_i and C_r , we have to calculate the correspondence of characteristics list (modeled as nodes) and, this, for each tumor included in the study. We apply, in this level; the Euclidian distance defines

$$\text{as follows: } d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (2)$$

Where n is the components number of the two vectors (characteristic's list) x and y .

Generally, the similarity function is applied on the tumor's membership vectors and described as follows:

$$\text{Sim}(\vec{I}, \vec{Q}) = \frac{\sum_{c=1}^C I_c * Q_c}{\sqrt{\sum_{c=1}^C I_c^2} * \sqrt{\sum_{c=1}^C Q_c^2}} \quad (3)$$

Where $\vec{I}$ and $\vec{Q}$ are the appartenance vectors to the tumors list of; successively; C_i and C_r .

We propose the following pseudo code to explain our method:

Similarity Measurement Algorithm.

Background:

A set of validated N Cases, C_i , $i = 1 \dots N$

A set of tumor classes $c = \{1, \dots, C\}$

Preprocess:

Draw up the membership vectors of the stored cases to the tumor's classes; we initialize a vector V_i for each case.

Input:

A MRI case with its semantic information's: C_r .

Distance calculation:

For a Query Case C_r :

- ✓ We determine the degree of membership of every state d_{jkY} of the descriptor d in the class c : d_{jkY} is defined as the average degree of the descriptor membership of the current class in the state d_{jk} .
- ✓ We deduct the membership degree of C_r to the available tumor's class c . This degree is the result node value of the network after inference. This task is repeated for all available classes, the score is added in V_r .
- ✓ We define the correspondence degree between C_r and C_i by the Cartesian product between both vectors V_r and V_i as follows:

$$S_{ri} = \sum_{c=1}^C V_{rc} \oplus V_{ic} \quad (5)$$
- ✓ The degree is, then, added in the final result vector.

Output:

A sorted vector with similarity score to case C_r .

C. Framework Model

The described methodology above is concretized by implementing an application to test results into simulations of real cases. The following figure describes the several components of the application:

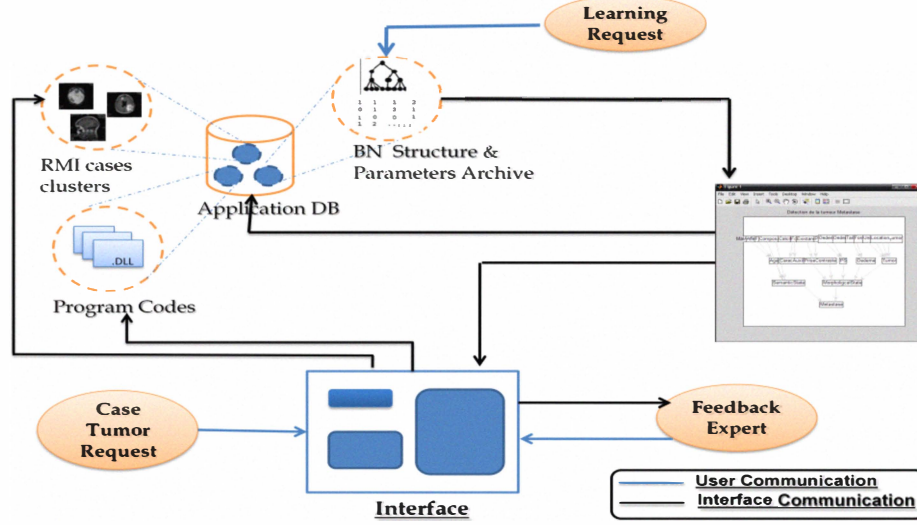


Figure 2. Architecture of the Bayesian MRI Cases Retrieval System

VI. EXPERIMENTS AND DISCUSSION

A. Experimental Data

All sample images are acquired from 1.5 Tesla “General Electric’s” MRI equipment. The examples data base consists of 56 samples collected from the radiology service in “Sahloul” hospital. Each examination can contain between 8 and 16 images selected by the experts. The already built base is heterogeneous and regroups all the tumors included and indicated above with different degrees of presence. The initial base is subdivided in two sub sets:

- A training base, made up of 46 cases,
- A test base, made up of 10 cases, to evaluate the efficiency and the precision of our contribution.

The algorithm evaluation is ensured by a diversified set of simulations. Experimental simulations aim to test the precision rate of the decision results of the implemented model.

Each test deck consists of three sub classes: a first simulation without training and with a distribution of the probabilities of the observed nodes proposed by the experts. Thus, we count a distribution for each concerned tumor (noted **Sim-Expert** in what follows). The second simulation is defined by applying complete data training, being the maximum Likelihood with the posterior algorithm (noted **Sim-Train1**). The maximum likelihood training algorithm considers the probability of an event as being its frequency of appearance:

$$\hat{P}(X_i = x_k | Pa(X_i) = x_j) = \frac{N_{i,j,k}}{\sum_k N_{i,j,k}} \quad (4)$$

Finally, a third simulation is applied with an exercise of incomplete data training method which is the EM algorithm [5] (noted **Sim-Train2**).

B. Results

We use two indices to evaluate the retrieval effectiveness of our proposition which are the mean sensitivity and the recall rate. The mean sensitivity (S) is the number of true

retrieved cases in a result window fixed as 5, in accordance with expert’s needs. It is defined as:

$$S = \text{relevant cases in top } N / N \quad (5)$$

Where N is the preset results window and it is fixed at 5.

The second index is the recall rate (R). It defines the number of relevant retrieved cases per the number of relevant cases in the data base. It is described as follows:

$$R = \text{number of relevant retrieved cases in top } N / \text{number of relevant cases in data base} \quad (6)$$

Thus, we present the performance indices of the system simulations. The first table schematizes the precision rate of the system results. The second one, from its part, represents the results system recall rate distribution. In both tables, each line represents a simulation which is marked by the tumor interpreted in the real diagnosis (the number of the tumor is referred to indicated table above):

TABLE II. THE SENSITIVITY MEAN OF THE SIMULATIONS WITH THREE APPLIED METHODS

	Sim-Expert	Sim-Train1	Sim-Train2
Simulation 1: (3)	0.8	0.6	0.4
Simulation 2: (1)	0.8	0.8	0.2
Simulation 3: (4)	0.6	0.4	0.6
Simulation 4: (3)	0.8	0.4	0.4
Simulation 5: (1)	0.8	0.4	0.2
Simulation 6: (2)	0.8	0.4	0.4
Simulation 7: (6)	0.4	0.2	0.4
Simulation 8: (3)	0.8	0.6	0.4
Simulation 9: (2)	0.8	0.4	0.2
Simulation 10: (5)	0.2	0.2	0.2
Mean Sensitivity Rate	68%	44%	34%

TABLE III. THE RECALL RATE OF THE SIMULATIONS WITH THREE APPLIED METHODS

	Sim-Expert	Sim-Train1	Sim-Train2
Simulation 1: (3)	0,18	0,13	0,09
Simulation 2: (1)	0,34	0,34	0,09
Simulation 3: (4)	0,43	0,29	0,43
Simulation 4: (3)	0,18	0,09	0,09
Simulation 5: (1)	0,34	0,17	0,09
Simulation 6: (2)	0,5	0,25	0,25
Simulation 7: (6)	0,4	0,2	0,4
Simulation 8: (3)	0,18	0,13	0,09
Simulation 9: (2)	0,5	0,25	0,12
Simulation 10: (5)	0,2	0,2	0,2
Mean Recall rate	32,5%	20,5%	18,5%

The following diagram shows us the performance variation of the applied methods:

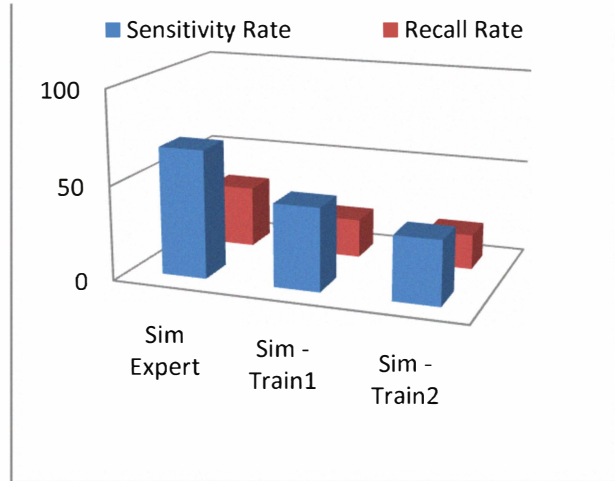


Figure 3. Simulations performance indices comparison

We have showing in table 2 the precision rate variations of the simulations desks. We can observe an obvious supremacy of simulations with the probabilities' distribution defined by the experts (Sim-Expert) which reaches 68%. Obviously, the two other simulations desks (with parameters learning initialization) marks the least precision rate results. At the same time, the same remark is deduced from the recall results variations. This stands to reason since the current data base still limit of cases number which affects directly the probabilities distribution nodes on tumor's network. Tumor's identification in the interpretation process is a hard task considering descriptors number included in the decision process and the close similarity between tumors description. At this level, we envisage perfecting system efficiency by performing our data base and propose a powerful similarity measurement algorithm.

VII. CONCLUSION

In this article, we have proposed a retrieval model for MRI cases containing cerebral tumors based on a Bayesian network model. We focused on the similarity algorithm and we have raised an experimentation section to test its performance. The proposed algorithm investigates the contribution of the probability reasoning in a bayesian network inference. At the time of an inference, the quantified similarity of a query case is measured according its distance to the cases recorded after measuring the position of each one to the tumors class. A test case is introduced as a signature including a vector of combining characteristics (visual characteristics and contextual information).

Work described in this paper can constitute a kernel for a complete cerebral tumors retrieval framework. We intend to focus on characteristic's extraction stage and couple it with similarity measurement module detailed in this paper.

REFERENCES

- [1] C. R. Shyu, A. M. Aisen et al., "ASSERT: A Physician-in-the-Loop Content-Based Retrieval System for HRCT Image Databases," Computer Vision and Image Understanding Vol. 75, Nos. 1/2, July/August, pp. 111–132, 1999.
- [2] C. Y. Lin, J. X. Yin, X. GAO, J-Y.Chen and P. Qin, "A Semantic Modeling Approach for Medical Image Semantic Retrieval Using Hybrid Bayesian Networks," proc. of the Sixth International Conference on Intelligent Systems Design and Applications, 2006, doi: 10.1109/ISDA.2006.253884.
- [3] D. M. Squire, " Learning a similarity-Based distance measure for image database organization from human partitionings of an image set," proc. of the Fourth IEEE Workshop on Applications of Computer Vision (WACV'98), Princeton, NJ, USA, pp. 88 – 93, Oct. 1998.
- [4] D. Wang and X. Ma, "A Hybird Image Retrieval System with User's Relevance Feedback Using Neurocomputing," Journal Informatica, Volume 29, pp. 271 – 279, 2005.
- [5] F. Dellaret. "The Expectation Maximization Algorithm," Technical Report number GIT-GVU-02-20, Feb. 2002.
- [6] G. Quéllec et al., "Multimodal medical case retrieval using Bayesian networks and the Dezert-Smarandache theory," Proc. 5th IEEE International Symposium on Biomedical Imaging: From Nano to Macro, May. 2008, pp. 245 – 248, doi: 10.1109/EMBS.2008.4649173.
- [7] H. Shao, W. Cui and H. Zhao, "Medical Image Retrival Based on Visual Contents and Text Information," Proc. 2004 IEEE International Conference on systems, Man, and Cybernetics, Oct. 2004.
- [8] I. J. Cox, M. L. Miller, T. P. Minka, T. V. Papathomas and P. N. Yianilos, "The Bayesian Image Retrieval System, PicHunter. Theory, implementation, and Psychophysical Experiments," IEEE Transactions On image processing, Vol 9, N°1, Jan. 2000
- [9] J. Bernardo, A. Smith, "Bayesian Theory," Wiley, 2000.
- [10] J. Cheng, R. D. Bell and W. Liu, "Learning Bayesian Networks from Data: An information-theory based approach," Artificial Intelligence, Vol. 137, pp 43 – 90, May. 2002.
- [11] J. Lio, A. E. Savakis and A. Singhal, "A Bayesian network – based framework for semantic image understanding," Pattern Recognition, Vol 34, pp 919 – 934, June 2005.
- [12] J.Stuart Russell and P. Norvig, "Artificial Intelligence: A Modern Approach," Pearson Education, Ichiro Satoh. Self-organizing multi-agent systems for data mining, 2003. 57.
- [13] K. J. Zulch, "Histological Typing of tumours of the Central Nervous System," Geneva: W.H.O. 66 pp. Sw Fr. 49, 1980.

- [14] L. G. Shapiro, I. Atmosukarto et al., "Similarity-Based Retrieval for Biomedical Applications," *Case-Based Reasoning Images and Signals*, Springer Berlin / Heidelberg, Vol. 73, 2008, p. 355-387, doi: 10.1007/978-3-540-73180-1.
- [15] P. Naim, P. Willemin, P. Leray, O. Pourret and A. Becker, *Les réseaux Bayesian Networks*, Eyrolles, Paris, 2004.
- [16] Q. Zhang and E. Izquierdo, "A Bayesian Network approach to multi – feature based Image Retrieval," *proc. of the International Conference on Semantic and Digital Media Technologies*, vol. 4036, pp. 138 – 157, Dec. 2006.
- [17] S. Dube, S. Elsaden, S. Cloughesy and T.F. Sinha, "Content Based Image Retrieval for MR Image Studies of Brain Tumors," *Proc of the 28th IEEE EMBS Annual International Conference (EMBS'06)*, Sept 2006, pp 3337 – 3340, doi: 10.1109/IEMBS.2006.260262.
- [18] T.M. Lehmann et al., "Content-based Image Retrieval in Medical Applications," *Methods Inf Med Journal*, Vol 43, pp 354 – 361, 2004.
- [19] Y. Liu et al., "Semantic Biomedical Image Indexing and Retrieval," *Proc. Trends and Advances in Content-Based Image and Video Retrieval*, Feb. 2004.
- [20] Y. Rui, T. S. Huang and S-F. Chang, "Image retrieval: Current techniques, promising directions, and open issues," *Journal of Visual Communication and Image Representation*, 10(1), 39–62, 1999.